



Multi-drug resistant (MDR), extended spectrum beta-lactamase (ESBL) producing and carbapenem resistant *Escherichia coli* in rescued Sloth bears (*Melursus ursinus*), India

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Abstract

The study reports the multi-drug resistant (MDR), extended spectrum beta-lactamase (ESBL) producing and carbapenem resistant *Escherichia coli* (CRE) isolated from rescued sloth bear (*Melursus ursinus*), India. Non-duplicate faecal samples from 21 adult rescued sloth bears were collected at once during 2015–2016 and processed for isolation of *E. coli* and anti-bacterial susceptibility pattern. From 21 samples, 45 *E. coli* were isolated and on phenotypic screening, 23 were MDR, 17 were ESBL producers, and five were carbapenem-resistant (CR). Three *E. coli* isolates (6.67%, 3/45) showed no resistance, however 42 isolates (93.33%, 42/45) exhibited resistant to at least one antibiotics. The MDR isolates carried beta-lactamase, chloramphenicol, aminoglycosides, tetracycline, fluoroquinolone, and sulphadimidine resistance genes. All the phenotypic ESBL producing isolates harbored *bla*CTX-M genes. On genotypic screening, three CRE (60.0%, 3/5) were positive for *bla*NDM carbapenemase gene and efflux pump-mediated carbapenem resistance was detected in two CRE isolates (40.0%, 2/5) which were negative for carbapenemase genes. The CRE isolates (n = 5) also co-harbored AMR genes like *bla*TEM-1, *bla*AmpC, *qnrA*, *qnrB*, *qnrS*, *tetA*, *tetB* and *sulI*. Virulence screening of the resistant isolates detected the presence of *Stx1* (n = 1), *Stx2* (n = 3), *eaeA* (n = 4) and *hlyA* (n = 3) genes. Plasmid incompatibility (Inc) typing revealed that two isolates harboured *bla*NDM-5 gene on Inc11 and one isolate on IncF plasmid. Apart from the NDM gene, the plasmids also carried tetracycline, beta-lactamase and quinolone resistance genes. The plasmid multilocus sequence typing (pMLST) of the *E. coli* Inc11 plasmid showed the Sequence Type (ST) 297. This appears to be the first report of MDR, ESBL producing and *bla*NDM-5 genes on Inc11 and IncF plasmids from rescued sloth bear.

Keywords Sloth bear · Carbapenem · ESBL · *E. coli* · Virulence

Introduction

Antimicrobial resistance is an emerging global threat and needs immediate attention. Genes conferring AMR can spread quickly by horizontal gene transfer to other non-resistant

bacteria and thereby disseminating AMR throughout microbial communities (Allen et al. 2010). Wild birds and mammals acts as important reservoirs and potential vectors for the spread of resistant bacteria and genetic determinants of AMR (Carroll et al. 2015). Wild animals are not directly exposed to antimicrobials; though, they may be infected with antibiotic-resistant bacteria via the acquisition of resistant bacteria from human sources, agricultural facilities and associated contaminated environments (Dolejska et al. 2007). Antimicrobial agents, such as antibiotics, are used extensively in human and veterinary medicine and often as part of routine agricultural production (Kummerer 2009). The prevalence level of resistant bacteria in the community is often linked with human activity (Bonnedahl et al. 2009; Alroy and Ellis 2011).

Extended spectrum beta-lactamases (ESBL) and/or carbapenemases producing *Escherichia coli* are of great

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concern worldwide and represent a complex multidimensional problem involving humans, animals and the environment (Atterby et al. 2017). Carbapenems are one of the most important groups of antimicrobials and are considered as a last line of defence for the treatment of severe infections in humans. Though carbapenem resistance in animals is rare, reports are available on carbapenem resistant *E. coli* in animals of India (Pruthvishree et al. 2017, 2018; Nirupama et al. 2018; Murugan et al. 2019; VinodhKumar et al. 2021).

Sloth bears are myrmecophagous and habitually seen as solitary although they are sometimes seen in groups. In India sloth bears are used in circus and by nomadic street performers (Kalandar community). Wildlife SOS, a not-for-profit organization rescue and rehabilitate these animals (Menezes 2021) and we hypothesize that, in these rehabilitation centers the close association of sloth bears among themselves and human may be a driving factor for transfer of resistant bacteria among them. The study was carried out to identify the molecular epidemiology of MDR, ESBL producing and carbapenem resistant *E. coli* isolated from rescued sloth bear. We selected *E. coli* for the AMR pattern in sloth bear since its ubiquitous nature in animals and humans and its easy and inexpensive isolation methods.

Materials and methods

Sample collection and isolation of *E. coli*

A cross-sectional study was conducted on the request of authorities of Agra Bear Rescue Facility (ABRF) for screening of antimicrobial resistance of *E. coli* in rescued sloth bears in India during 2015–16. During the study period the rehabilitation center caged 190 sloth bears and approximately 10% of the healthy animals were randomly selected for bacterial resistance screening. The fecal samples were directly collected from rectum of each sloth bear ($n = 21$) during the routine health checkup performed under general anesthesia. Non-duplicate samples from apparently healthy adult sloth bears were collected at single time point in sterile transport swabs (HiCulture, HiMedia, India). Following enrichment in MacConkey broth at 37°C for 6–10 h, a loop full of enriched broth was streaked on MacConkey agar and incubated at 37 °C for 12–18 h. Based on colony morphology, two to three lactose-fermenting colonies selected were streaked on eosin methylene blue (EMB) agar medium and incubated at 37 °C overnight for preliminary characterization. Then, the colonies showing characteristic metallic sheen on EMB agar were selected for oxidase, Voges-Proskauer, methyl red, indole, citrate, urea hydrolysis, catalase, casein hydrolysis, nitrate reduction, and sugar fermentation biochemical tests to confirm their identity. The *E. coli* reference strains used in this study were retrieved

from the Division of Epidemiology, ICAR-Indian Veterinary Research Institute, Izatnagar repository (GenBank accession number: KU318701, KU318691, KU382501).

Antimicrobial susceptibility testing

The biochemically confirmed *E. coli* ($n = 45$) isolates were tested for antibiotic susceptibility pattern with amoxicillin (10 µg), amoxicillin/clavulanic acid (20/10 µg), aztreonam (30 µg), chloramphenicol (30 µg), ceftriaxone (30 µg), cefazolin (30 µg), cefpodoxime (10 µg), ceftazidime (30 µg), cefotaxime (30 µg), cefepime (30 µg), cefixime (5 µg), cefoxitin (30 µg), piperacillin—tazobactam (100/10 µg), cefotaxime/clavulanic acid (30/10 µg), tetracycline (30 µg), nitrofurantoin (300 µg), gentamicin (10 µg), kanamycin (30 µg), co-trimoxazole (25 µg), ciprofloxacin (5 µg) and norfloxacin (10 µg) by disc diffusion method (Hudzicki 2009). The CLSI (2014) breakpoints were used for interpretation of susceptibility patterns. For phenotypic identification of ESBL producers, the combination disc method using cefotaxime and ceftazidime with and without clavulanic acid was used (Andrews 2003) and the isolates were further confirmed by ESBL E-test (HiMedia, Mumbai, India). Screening for carbapenem resistance was also carried out using the disc diffusion method with meropenem (MEM, 10 µg), imipenem (IMP, 10 µg), and ertapenem (ETP, 10 µg) disks (BD, Sparks, MD, USA) along with ethylene diamine tetra acetic acid (EDTA 1900 mg/disc) for screening of metallo beta-lactamase (MBL) production (Pruthvishree et al. 2017). A keyhole reaction between carbapenem drug and EDTA was considered positive for metallo beta-lactamase (MBL) production. The minimum inhibitory concentration (MIC) for ceftriaxone, cefotaxime and meropenem was determined using E-test strip (HiMedia, Mumbai, India).

Genotypic screening for antimicrobial resistance (AMR)

Phenotypically resistant *E. coli* isolates were further analyzed for AMR encoding genes. The genomic DNA was extracted from *E. coli* isolates using a QIAamp® DNA Mini Kit (QIAGEN, Hilden, Germany) and screened for the presence of antimicrobial resistance genes (ARG), including beta-lactamase, major carbapenemases, sulphonamide, quinolone and tetracycline resistance genes. The isolates were also screened for *Stx1*, *Stx2*, *eaeA* and *hlyA* virulence genes (Paton and Paton 1998) in order to find the resistant isolates carrying the virulence genes. The PCR reactions were optimized individually in 25 µL volumes, with 10 pmol of each individual primer used (supplementary Table 1). The PCR amplicon from all reactions was electrophoresed in 0.5 X TBE buffer using 1.5% agarose gel and visualized under UV illumination (Syngene, USA). Representative

PCR amplicons were purified and sequenced by a commercial sequencing services (Eurofins, Bengaluru, India). The sequencing results were obtained, and similarity searches were performed using the BLAST algorithm (<https://blast.ncbi.nlm.nih.gov/Blast.cgi>). The sequence data were submitted to GenBank, and accession numbers were obtained.

Plasmid replicon and multilocus sequence typing

Plasmid DNA was isolated using a QIA prep Spin miniprep kit (QIAGEN, Helsinki, Finland) from the *bla*NDM positive *E. coli* isolates and PCR based replicon typing (PBRT) was used to determine the plasmid incompatibility groups (Johnson et al. 2007). Plasmid multi locus sequence typing (MLST) was performed using specific primers (<https://pubmlst.org/plasmid/primers/>) and sequenced by the Sanger method by a commercial sequencing service (Eurofins India Ltd, Bengaluru). The sequences were edited using BioEdit v7.0.5 and queried in the plasmid MLST website (https://pubmlst.org/bigdb?db=pubmlst_plasmid_seqdef) for an allelic profile, sequence type and clonal complex.

Efflux pump screening

The *E. coli* isolates not showing carbapenemase gene mediated resistance were screened for the presence of active efflux pump by the ethidium bromide cartwheel method in order to find the non-carbapenemase mediated carbapenem resistance (Martins et al. 2013).

Statistical analysis

Antibiotic susceptibility profiles and gene occurrence was converted into binary code. For each antibiotic, 0 denoted susceptibility and “1” denoted resistance. The intermediate resistant isolates were included in resistant category. Similarly, the absence and presence of a resistant gene was denoted as “0” and “1” respectively. Principal component analysis (PCA) was performed using the package Factoextra (version 1.0.6) in R software (version 3.6.1).

Results and discussion

A total of 45 *E. coli* were isolated from 21 sloth bear faecal samples and the antibiograms showed that the isolates were resistant to gentamicin (62.2%, 28/45), cefotaxime (60.0%, 27/45), ceftazidime (60.0%, 27/45), cefotaxime/clavulanic acid (57.8%, 26/45), ceftriaxone (55.6%, 25/45), ceftazidime/clavulanic acid (55.6%, 25/45), norfloxacin (55.6%, 25/45), cefpodoxime (53.3%, 24/45), cefoxitin (53.3%, 24/45), piperacillin/tazobactam (53.3%, 24/45), cefexime (51.1%, 23/45), ciprofloxacin (51.1%, 23/45), tetracycline

(51.1%, 23/45), sulphadimidine/trimethoprim combination (46.7%, 21/45), cefepime (46.7%, 21/45), chloramphenicol (42.2%, 19/45), nitrofurantoin (40.0%, 18/45), aztreonam (35.6%, 16/45), meropenem (11.1%, 5/45), imipenem (8.9%, 4/45) and ertapenem (8.9%, 4/45). Of the 45 isolates, 51.1% (23/45) were resistant to more than two classes of screened antibiotics and classified as MDR. Three (6.67%, 3/45) and 42 (93.33%, 42/45) *E. coli* isolates showed no resistance and resistance to at least one antibiotics screened in this study, respectively.

The ESBL screening by combined disc method revealed 37.8% (17/45) isolates as ESBL producers (Table 1). Similar kind of antibiotic resistance pattern was documented in *E. coli* isolated from captive blackbuck and leopard of India (VinodhKumar et al. 2021). Comparable to this study, many authors documented multidrug resistant and ESBL producing *E. coli* in wild animals (Costa et al. 2006; VinodhKumar et al. 2021). Literature reports indicate that wild birds are reservoirs of extended-spectrum β -lactamase (ESBL) and AmpC β -lactamase producing *E. coli* (Alcala et al. 2016). Interestingly, a significant percentage of MDR *E. coli* isolates was detected in wild birds and animals such as black-headed gulls (3.3%), starlings (19.2%), herring gulls (3.3%), deer (3.3%) of Ireland (Carroll et al. 2015). In contrast, the present study documented MDR and ESBL producing *E. coli* in rescued sloth bears.

Screening for carbapenem resistance showed that 11.11% (5/45) isolates were resistant to carbapenem drugs and the five carbapenem resistant isolates were MDR and ESBL producers. Interestingly, three carbapenem-resistant isolates showed a keyhole reaction between EDTA and carbapenem drugs, indicating metallo beta-lactamase production (MBL). Metallo-carbapenemases are versatile in hydrolyzing penicillins, cephalosporins, monobactams, and carbapenems (Queenan and Bush 2007). The carbapenem resistant, metallo beta-lactamase producing *E. coli* isolates were reported in piglets, dog, blackbuck and leopard in India (Pruthivishree et al. 2017, 2018; VinodhKumar et al. 2021).

The phenotypically resistant isolates were analyzed for AMR encoding genes. All the ESBL producing isolates were MDR *E. coli* and genetic screening of the ESBL positive isolates (n = 17) revealed the presence of ESBL encoding genes such as *bla* CTX-M-1 (n = 10), *bla* CTX-M-15 (n = 7) and other antibiotic resistance genes like *bla* TEM (n = 6), *bla* AmpC (n = 4), *qnr* S (n = 6), *qnr* A (n = 3), *qnr* B (n = 3), *sul* I (n = 7), *sul* II (n = 3), *tet* A (n = 4), *tet* B (n = 5). Similarly, most of the ESBL producing *E. coli* isolates in wild birds were also multi-drug resistant and the most common resistant phenotypes were beta-lactams, quinolones, tetracycline, and sulfamethoxazole/trimethoprim (Alcala et al. 2016). The most common ESBL gene in dog, cattle and pig was *bla* CTX-M-1, while in humans *bla* CTX-M-14 and *bla* CTX-M-15 were the dominant genes. However, recent studies

Table 1 Antibiotic susceptibility pattern, virulence and antibiotic resistance determinants of *E. coli* isolates (n=45) from rescued sloth bear

S. no	Isolates	Antibacterial resistance pattern																			Minimum inhibitory concentration (µg/mL)	Virulence and antibiotic resistance determinants	Accession number						
		AMX	AMC	CZ	ATZ	COT	CX	CTX	CFM	CPD	CIP	CAZ	CTR	K	C	GEN	TE	FOX	F/M	TZP				NX	IMP	MRP	ERP	CTX	CTR
1	SB-1	S	R	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	nd	nd	nd	-	nd
2	SB-2 ESBL	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	S	S	8	12	nd	CTX-M-1, qnrA, AmpC, catI, sulI, tetA, ac (3)-IV, eaeA	nd
3	SB-3 ESBL	R	R	R	S	R	R	R	R	R	R	R	R	R	S	R	R	R	S	R	R	S	S	S	12	18	nd		nd
4	SB-4	R	S	R	S	R	R	S	S	R	S	S	S	R	S	S	S	S	S	S	R	S	S	S	nd	nd	nd	-	nd
5	SB-5	R	S	R	S	S	R	S	S	S	S	S	S	R	R	R	S	S	S	S	S	S	S	S	nd	nd	nd	floR	nd
6	SB-6	R	R	R	S	R	R	S	S	S	S	S	S	R	S	S	S	R	S	S	S	S	S	S	nd	nd	nd	Stx2	nd
7	SB-7	R	R	R	S	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	S	S	S	12	16	nd	CTX-M-15, qnrS, Stx2	MK559368
8	SB-8ESBL	R	R	R	R	R	R	R	R	R	R	R	R	S	R	R	R	R	S	R	S	S	S	S	8	12	nd	AmpC, sulI, floR, ac (3)-IV	nd
9	SB-9	R	S	S	S	R	R	S	R	S	S	R	S	R	R	S	S	S	S	R	R	S	S	S	nd	nd	nd	floR, TEM-1	nd
10	SB-10	R	S	S	S	R	S	R	S	S	S	R	S	R	R	R	S	S	S	S	R	S	S	S	nd	nd	nd	floR	nd
11	SB-11	R	S	S	S	S	S	R	S	S	S	R	S	R	R	S	S	S	S	S	S	S	S	S	nd	nd	nd	floR	nd
12	SB-12 ESBL	R	R	R	R	R	R	R	R	R	R	R	R	R	S	R	R	R	R	R	R	S	S	S	12	16	nd	CTX-M, qnrS, ac (3)-IV, eae A	nd
13	SB-13 ESBL	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	S	S	S	32	64	nd	floR, dhfrI	nd
14	SB-14	R	S	R	S	S	S	S	S	S	S	S	S	S	R	S	S	S	R	S	R	S	S	S	nd	nd	nd	catI	nd
15	SB-15ESBL	R	R	R	S	R	R	S	R	R	R	R	R	R	R	R	R	R	R	R	R	S	S	S	16	32	nd	CTX-M, qnrA, sulI, floR	nd
16	SB-16 ESBL	R	R	R	S	R	R	S	R	R	R	R	R	R	S	R	R	R	R	R	R	S	S	S	8	12	nd	CTX-M, qnrB, tetA, hlyA	nd
17	SB-17 ESBL	R	R	R	R	R	R	R	R	R	R	R	R	R	S	R	R	R	R	R	R	S	S	S	12	16	nd	CTX-M, qnrS, tetB, dhfrI, TEM-1	MK559374
18	SB-18	R	R	S	S	S	S	S	S	S	S	S	S	R	R	S	S	S	S	S	S	S	S	S	nd	nd	nd	catI	nd
19	SB-19ESBL	R	R	R	S	R	R	S	R	R	R	R	R	S	R	R	R	R	R	R	R	S	S	S	8	12	nd	CTX-M, qnrB, sulII, catI, ac (3)-IV	nd
20	SB-20 ESBL	R	R	R	R	R	R	R	R	R	R	R	R	R	S	R	S	R	S	R	R	S	S	S	16	18	nd	CTX-M, sulI, dhfrI	nd
21	SB-22 ESBL	R	R	R	S	R	R	R	R	R	R	R	R	R	S	R	R	R	S	R	S	S	S	S	12	16	nd	CTX-M, qnrA, TEM-1	nd
22	SB-23 ESBL	R	R	R	S	R	R	R	R	R	R	R	R	S	R	R	R	R	R	R	R	S	S	S	32	64	nd	CTX-M, TEM-1, qnrS, sulI, AmpC, tetB, catI, ac (3)-IV, dhfrI	MK559371, MK559373
23	SB-24	R	S	S	S	S	S	S	S	S	S	S	R	S	S	S	S	S	S	S	S	S	S	S	nd	nd	nd	hlyA	nd
24	SB-25	S	R	S	S	S	S	R	S	R	S	S	R	S	R	R	R	R	S	S	S	S	S	S	nd	nd	nd	ac (3)-IV	nd
25	SB-26	R	S	S	S	S	S	S	S	S	S	S	S	R	S	S	S	S	S	S	S	S	S	S	nd	nd	nd	-	nd
26	SB-27/17-CRE	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	8	16	16	NDM-5, TEM-1, qnrS, sulII, catI, ac (3)-IV	MF136747, MK559369, MK559372

Table 1 (continued)

S. no	Isolates	Antibacterial resistance pattern																Minimum inhibitory concentration (µg/mL)				Virulence and antibiotic resistance determinants	Accession number							
		Antibacterial resistance pattern																Minimum inhibitory concentration (µg/mL)												
		AMX	AMC	CZ	ATZ	COT	CX	CTX	CFM	CPD	CIP	CAZ	CTR	K	C	GEN	TE	FOX	F/M	T/ZP	NX			IMP	MRP	ERP	CTX	CTR	MRP	
27	SB28/ 12-CRE	R	R	S	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	16	32	32	NDM-5, CTX- M,TEM-1,AmpC	MK559365, MK559367, MK559370, MK559375	
28	SB28/ 16-CRE	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	16	8	8	NDM-5, CTX-M, Sxx1,tetB,cat1,aph (3')-IIa	MK559366	
29	SB- 29/18-CRE	R	R	R	S	S	R	R	R	R	R	R	R	R	R	S	R	R	R	R	R	R	R	R	S	8	32	16	tetB	Efflux pump
30	SB-30 CRE	R	S	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	16	16	32	CTX-M-qnrS,floR, dhfr1,sxx2	Efflux pump	
31	SB-31 ESBL	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	S	R	R	R	R	R	R	S	16	32	nd	CTX-M,sul1,tetB, dhfr1	nd	
32	SB-32 ESBL	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	S	8	8	nd	CTX-M,qnrB,cat1	nd	
33	SB-33 ESBL	R	R	S	S	S	R	R	R	R	S	R	R	S	R	R	S	R	R	S	R	R	S	S	32	32	nd	CTX-M,tet A, eaeA	nd	
34	SB-34 ESBL	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	S	S	16	16	nd	CTX-M,sul1,tetA, aph (3')-IIa	nd	
35	SB-35	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	nd	nd	nd	sulII	nd	
36	SB-36	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	nd	nd	nd	-	nd	
37	SB-37	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	R	R	S	S	S	nd	nd	nd	-	nd	
38	SB-38	S	S	S	S	S	S	S	S	S	S	S	S	S	R	S	S	R	S	S	S	S	S	S	nd	nd	nd	aph (3')-IIa	nd	
39	SB-39	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	nd	nd	nd	-	nd	
40	SB-40	S	S	S	S	S	S	R	S	S	S	S	S	R	S	S	R	S	S	S	S	R	S	S	nd	nd	nd	ac (3)-IV	nd	
41	SB-41	S	R	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	nd	nd	nd	hlyA	nd	
42	SB-42	S	S	S	S	S	S	S	S	S	S	S	S	R	R	S	S	S	S	S	S	S	S	S	nd	nd	nd	eaeA	nd	
43	SB-43	S	S	S	S	S	S	S	S	S	S	S	S	S	S	R	S	S	S	S	S	S	S	S	nd	nd	nd	aph (3')-IIa	nd	
44	SB-44	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	R	S	nd	nd	nd	-	nd	
45	SB-45	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	S	nd	nd	nd	-	nd	

R resistant, S sensitive, AMX amoxicillin, AMC amoxicillin-clavulanic acid, ATZ aztreonam, C chloramphenicol, CTR ceftriaxone, CPD cefepime, CAZ ceftazidime, CZ cefazolin, CFM cefixime, CTX cefotaxime, CPM cefepime, CIP ciprofloxacin, COT co-trimoxazole, ERP ertapenem, F/M nitrofurantoin, FOX cefoxitin, GEN gentamicin, IMP imipenem, K kanamycin, NX norfloxacin, MRP meropenem, TE tetracycline, TZP piperacillin-tazobactam, nd not done

have identified CTX-M-1 genes in humans (Madec et al. 2015) and rare identification of CTX-M-15 in pet animals (Dahmen et al. 2013). In the present study, *E. coli* isolated from sloth bear harbored both CTX-M-1 and CTX-M-15 variants indicating sharing of dominant ESBL gene pool among *E. coli* (Zong et al. 2008).

Screening for carbapenemase genes revealed that three isolates which showed phenotypic MBL production were positive for the *bla*NDM-5 gene. The *bla* NDM positive isolates also co-harboured *bla*CTX-M1/15, *bla*TEM-1, *qnrS*, *qnrA*, *qnrB*, *sul1*, *tetA* and *tetB* genes. Two CRE isolates (SB28/ 16-CRE, SB-30 CRE) carried both the virulence and antibiotic resistant genes. Similarly, carbapenem resistant *E. coli* carrying *bla*NDM and virulence genes were reported in captive leopards of India (VinodhKumar et al. 2021).

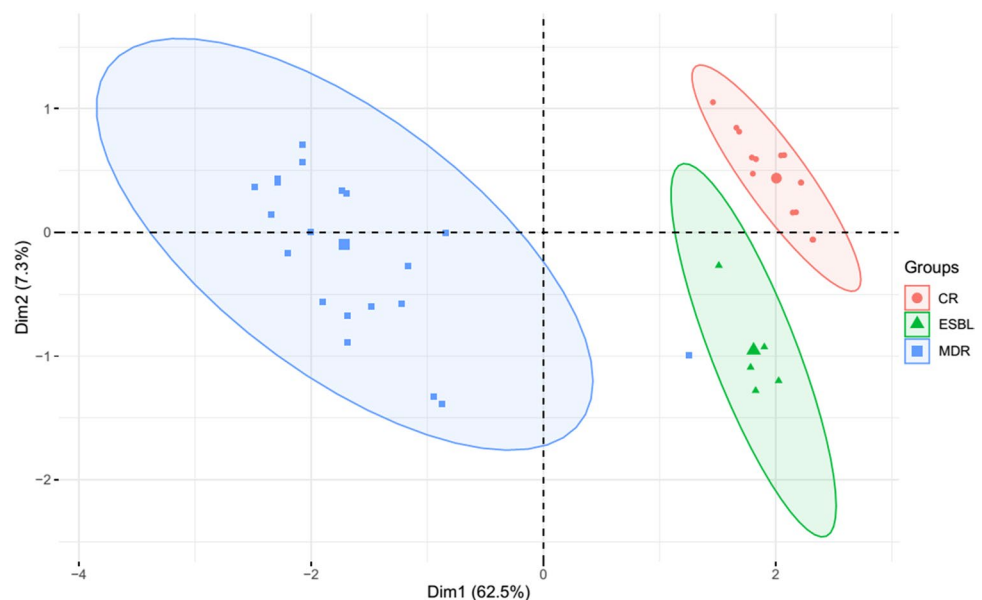
The efflux pump-mediated carbapenem resistance was detected in two (40%, 2/5) isolates. Similarly, the efflux pump mediated carbapenem resistance was noticed in *E. coli* isolated from piglets and dairy calves in India (Pruthvishree et al. 2017; Murugan et al. 2019). Chloramphenicol (*catI*, *floR*), aminoglycoside (ac (3)-IV, aph (3')-IIa), quinolone (*qnrS*, *qnrA*, *qnrB*), trimethoprim (*dhfrI*), sulphadimidine (*sul I*, *sul II*) and tetracycline resistance genes (*tetA*, *tetB*) were commonly detected in the phenotypic resistant isolates. Small free-living mammals' fecal samples harbored trimethoprim, ampicillin, ciprofloxacin, and cefotaxime resistant *E. coli* (Furness et al. 2017). Screening for virulence genes revealed that four ESBL producing (SB-3 ESBL, SB-12 ESBL, SB-16 ESBL, SB-33 ESBL) and two carbapenem-resistant isolate (SB28/ 16-CRE, SB-30 CRE) harbored any of the virulence genes screened (Table 1). In the same way, reports are available on ESBL producing and carbapenem resistant *E. coli* harboring virulence encoding genes

in piglets (Pruthvishree et al. 2017, 2018; VinodhKumar et al. 2019, 2021).

Plasmid replicon typing of the three *bla*NDM-5 positive *E. coli* revealed that the NDM gene was on the Inc11 plasmid of two isolates (SB27/ 17-CRE, SB28/ 12-CRE) and on the IncF plasmid of one isolate (SB28/ 16-CRE). The Inc11 and IncF plasmids also co-harbored quinolone, extended-spectrum cephalosporin, sulphonamide and tetracycline resistance genes such as *bla*CTX-M-15, *bla*TEM-1, *tetA*, *sul I* and *qnrA*. In the same way, IncF plasmids have been reported worldwide among *Enterobacteriaceae* and are one of the most prevalent incompatibility types involved in the transfer of resistance determinants (Yang et al. 2015). Recently, the IncF plasmid carrying multi-drug resistant genes were reported in *Shigella flexneri* isolates from India (Sethuvel et al. 2019). An earlier study also documented the plasmids carrying multiple antibiotic genes in Gram negative bacteria (Ramirez et al. 2014). Similarly, the IncF plasmid encoding beta-lactamase (NDM-1, OXA-1), aminoglycoside (*aacA4*, *aadA2* and *aacC2*), and extended-spectrum cephalosporin resistance (CTX-M-15) genes in *E. coli* ST131 isolated from humans was reported (Bonnin et al. 2012). Plasmids carrying the AMR genes have the potential to transfer the AMR determinants to non-resistant bacteria (Allen et al. 2010; Oladeinde et al. 2019).

The plasmid multilocus sequence typing (pMLST) revealed that the *bla*NDM positive *E. coli* belonged to ST297. Similarly, a recent study in India documented *bla*VIM gene in *E. coli* isolated from calf belonging to ST 297 (Murugan et al. 2019). Correspondingly, in a recent publication we reported Inc11 plasmid carrying *bla*NDM gene in *E. coli* isolated from captive leopard (VinodhKumar et al. 2021).

Fig. 1 Principal Component Analysis (PCA) plots of the drug resistant *E. coli* isolates. The *E. coli* are encompassed in 95% confidence intervals grouping based on the resistant group [Carbapenem (CR), Extended Beta-lactam (ESBL), Multidrug resistant (MDR)]



The ESBL producing, carbapenem and multidrug resistant *E. coli* were divided into three groups in the principal component analysis (PCA) and confirmed the difference in resistant pattern of the MDR, ESBL producing and carbapenem resistant *E. coli*. The *E. coli* were grouped in to three categories based upon their resistance pattern to antibiotics. The first two dimensions included in PCA map showed the most variance of data (first: 62.5%; second: 7.3%). The third and fourth dimensions, despite being significant, were not plotted due to the complexity of the corresponding output (Fig. 1). Similarly, PCA analysis of ESBL producing *E. coli* isolates were carried out in piglets and pig farm workers indicated a common phenotypic and genotypic resistance pattern among the piglet and the farm workers (Tamta et al. 2020).

In conclusion, the study documented the presence of MDR, ESBL producing and carbapenem resistant *E. coli* carrying virulence genes from rescued sloth bears in India. The study also witnessed few *E. coli* isolates co-harboured antimicrobial resistant and virulence genes. We need to focus on these isolates because the use of antibiotics possibly will select for *E. coli* carrying virulence genes and could accelerate the spread of virulence genes within *E. coli* populations. Hence, the *E. coli* isolates harboring antibiotic resistant with virulence genes are critically important in public health. The sloth bears may be a potential source for spread of these resistant bacteria to its handlers and immediate environment. To the best of our knowledge, this is the first report of IncF and Inc11 type plasmids carrying multidrug-resistant and NDM carbapenemase producing genes in *E. coli* from sloth bear.

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Declarations

Consent to publish We give consent to publish.

Conflict of interests The authors declare no conflict of interest.

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